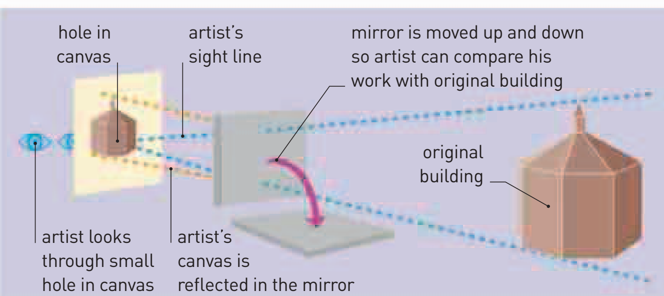
Italian artist Masolino da Panicale was an early master of visual perspective. He made good use of the technique in *St. Peter Curing a Cripple and the Raising of Tabitha*, painted for the Brancacci Chapel , Florence.

THE PRINCIPLES OF LINEAR PERSPECTIVE were known to the ancient Greeks, particularly **Euclid** who wrote of it in his *Elements*, but knowledge of these was lost after the fall of the Roman Empire.

Although Italian artist **Giotto** (1266–1337) had attempted to use algebraic formulas to create perspective, he had only partially succeeded. Renewed efforts to achieve true linear perspective included works from 1377 to 1397 by Italian mathematician **Biagio Pelacani** (*c.*1347–1416), who showed how **mirrors** could be used as **aids to view objects at a distance**. In 1415–16, Italian architect **Filippo Brunelleschi** (1377–1446) demonstrated in public for the first time the use of mirrors, by **reflecting an image of the Florence Baptistery** onto a 12 in- (30 cm-) canvas, which could then be drawn in perspective.

138 FEET THE WIDTH OF THE DOME OF FLORENCE CATHEDRAL



BRUNELLESCHI'S PERSPECTIVE

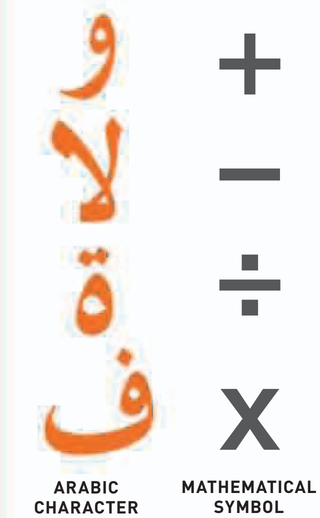
Florence architect and artist Filippo Brunelleschi used mirrors to recreate an accurate depiction of Florence's Baptistry on canvas. He realized that linear perspective could be used to give an accurate impression of a three-dimensional object on a two-dimensional surface. Using a single perspective point (a hole in the canvas) and a mirror, he produced a painting that was identical to the original.

Brunelleschi may have been taught the technique by Florentine physician **Paola Toscanelli** (1397–1482), but he did not publish his theories until 1460. The first full account of the **application of linear perspective to painting**, including the creation of a grid to organize the placement of objects in a picture and of the principles of the **vanishing point and horizon line**, was set out by **Leon Battista Alberti** (1404–72), another of Toscanelli's pupils, in his *On Painting* in 1436.

As well as being an innovator in drawing, Brunelleschi devised advanced machinery for the construction of his many building projects in Florence. Among these was a *colla grande* (great crane), **a massive barge-based hoist** that could lift weights of more than one ton, had three different lifting velocities, and could operate in reverse without unhitching the load. In 1421, the authorities in Florence granted him **the first recorded monopoly patent**. The Venetian government would go on to regularize the process of granting patents, giving inventors 10 years' monopoly rights, as long as the invention was properly registered.

The dome of Santa Maria del Fiore cathedral, Florence, is 137.8 ft (42 m) in diameter and 171 ft (54 m) high and took its architect, Brunelleschi, 16 years to complete.

IN 1420, THE MONGOL RULER ULUGH BEG (1411–49) had established a scientific institute at Samarkand, Uzbekistan; in 1424, he **started to build an observatory** there. It had a huge sextant that had a radius of 131 ft (40 m). Among the astronomers recruited was **Jamshid al-Kashi** (*c.*1380–1429), who produced **a mathematical encyclopedia** with a section on astronomical calculations, calculated the **value of pi** to 17 decimal places, and helped produce an extremely **accurate set of trigonometric tables**. In 1437, the astronomers at the observatory published the *Zij-i-Sultani*, **a star catalog** showing the position of 1,018 fixed stars.



A new mathematical language
The mathematician al-Qalasadi used short Arabic words for algebraic operations, such as wa ("and") for addition and ala ("over") for division.

Between 1430 and 1440, **Ibn Ali al-Qalasadi** (1412–86), a Spanish Muslim mathematician, published a work in which he used a series of **short words and abbreviations** to stand for **arithmetical operations in algebraic equations**. He was not the first to do so—such Arabic abbreviations had appeared a century earlier in North Africa, and Diophantus had devised a form of algebraic notation—but al-Qalasadi's widely diffused works were responsible for popularizing the system.

In 1436, Brunelleschi finally completed the dome of Florence Cathedral after 16 years work. The dome was **the largest unsupported structure yet built**, and Brunelleschi solved the problem of its weight by building a lighter inner shell, on which was built a tougher outer dome. He used a ring and rib pattern of stone and timber supports between the two shells, and devised a herringbone pattern for the bricks, both of which helped diffuse the weight of the structure.

Nicholas of Cusa (1401–64), a German philosopher, wrote a number of treatises such as *De Docta Ignorantia* (*On Learned Ignorance*), which included advanced astronomical and

1415–16 Brunelleschi demonstrates the use of **perspective** in Florence

1420 Ulugh Beg establishes **scientific institute** in Samarkand

1421 First recorded **patent granted to Brunelleschi** in Florence

1424 Ulugh Beg starts to build **observatory** at Samarkand

c.1425 Al-Kashi calculates the **value of pi** to 17 decimal places